Machine Learning & Python

Lecture Notes

Lecture 6 — Python Loops (04/09/2006)

Topics Covered

- What is Stable Diffusion? (What, Why, How)
- Loops in Python

Practical: Loops on Data Structures

→ Tuple
→ List
→ Set
→ Dictionary

Lecture 11 — Data Preprocessing

1. Structured Data (Rows and Columns)

Ordinal Data: Categories with a ranking/order (e.g., Low, Medium, High)
Nominal Data: Categories with no order (e.g., Name, Red, Blue, Green)
Numerical Data: Numeric values (e.g., 1, 3, 4, 5, 6, 7, 8)

2. Inconsistent Data

- Missing data
- Incomplete data

3. Data Reduction

Used to reduce the size of data while retaining the most important information.

4. Data Transformation

A step in data preprocessing (used in machine learning) where data is converted into a suitable format so a model can understand and use it effectively.

Types of Machine Learning

Supervised Learning — Learning with labeled data
Unsupervised Learning — Discovering patterns in unlabeled data
Reinforcement Learning — Learning to act based on feedback/reward

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Lecture 12 — Exploratory Data Analysis (EDA)

Note: The most common application of EDA is on structured (tabular) data. EDA is performed ***before*** *model training.*

Steps in EDA

1. **Data Cleaning** — Handle missing data, inconsistencies, etc.
2. **Descriptive Statistics** — Summary of large datasets
3. **Data Visualization**
4. **Data Distribution**
5. **Correlation Analysis**
6. **Outlier Detection**
7. **Data Transformation**

Descriptive Statistics

The branch of statistics that focuses on **summarizing and describing** the main features/attributes/variables of a dataset.

Measures of Central Tendency

- **Mean**: Average of all values
- **Median**: The center (middle) value
- **Mode**: The most frequently occurring value

Quartiles and IQR

Note: Quartiles divide a dataset into **four equal parts**, each representing 25% of the data. Applicable on **numerical data only** (not ordinal or nominal).

Q1 (First Quartile): The value below which 25% of the data falls.

Q2 (Second Quartile / Median): The value below which 50% of the data falls.

Q3 (Third Quartile): The value below which 75% of the data falls.

IQR (Interquartile Range): IQR = Q3 - Q1. Used to measure dispersion/spread.

Lecture 13 — Quartiles & IQR: Practical Example

Step-by-Step: Applying Quartiles on Salary Data

To understand the quartile concept clearly, we take **10 salary values** and apply the formula step by step.

Dataset (10 Salaries — already sorted in ascending order):

#	Salary	Position
1	200k	Lower half
2	250k	Lower half
3	300k	Lower half → Q1
4	370k	Lower half
5	410k	Lower half

6	450k	Upper half
7	500k	Upper half
8	550k	Upper half → Q3
9	590k	Upper half
10	620k	Upper half

Step-by-Step Formula Application

Step 1 — Sort the Data

Arrange all values in ascending order (already done above).

Step 2 — Find Q2 — Median

With 10 values, the median lies between the 5th and 6th values.

Q2 = (410 + 450) ÷ 2 = 860 ÷ 2 = 430k

Step 3 — Find Q1 — Lower Quartile

Take the lower half: 200, 250, 300, 370, 410 (5 values).

Q1 = middle value = 300k

Step 4 — Find Q3 — Upper Quartile

Take the upper half: 450, 500, 550, 590, 620 (5 values).

Q3 = middle value = 550k

Step 5 — Calculate IQR

IQR = Q3 – Q1 = 550 – 300 = 250k

This means the middle 50% of salaries are spread across a 250k range.

Step 6 — Detect Outliers using the 1.5 × IQR Rule

Lower Fence = Q1 – (1.5 × IQR) = 300 – 375 = –75k

Upper Fence = Q3 + (1.5 × IQR) = 550 + 375 = 925k

Any salary below –75k or above 925k would be an outlier.

Result: All 10 salaries are within range — NO outliers detected.

Summary of Results

Measure	Formula	Result
Q1 (First Quartile)	Median of lower half	300k
Q2 (Median)	(5th + 6th) ÷ 2	430k
Q3 (Third Quartile)	Median of upper half	550k
IQR	**Q3 – Q1**	**250k**
Lower Fence	Q1 – 1.5 × IQR	–75k

Upper Fence	Q3 + 1.5 × IQR	925k
Outliers	Values outside fences	None

Key Takeaway: IQR measures the **spread of the middle 50%** of data. It is resistant to extreme values (outliers), making it more reliable than the full range for understanding data dispersion.

Important Notes / To Explore

The following topics are flagged as important for further study:

✓ Hugging Face

✓ Pattern Identification (Lec 12 BML)

✓ Conventional Development (Lec 12 BML)